

EX.NO.: 08

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IMPLEMENTING POSITIONAL EMBEDDING

AIM

Implementing positional embeddings (sinusoidal and learnable) for a sequence of tokens in a transformer model.

DATASET DESCRIPTION

This code does not use a specific dataset but generates embeddings for a sequence of 6 tokens with an embedding size of 8.

TECHNIQUES USED

Sinusoidal positional encoding (deterministic) and learnable positional encoding (randomly initialized trainable parameters).

CODE AND OUTPUT

```
import torch
import numpy as np

def sinusoidal_positional_encoding(seq_len, d_model):
    positions = np.arange(seq_len)[:, np.newaxis]
    div_term = np.exp(np.arange(0, d_model, 2) * -(np.log(10000.0) / d_model))

    pos_enc = np.zeros((seq_len, d_model))
    pos_enc[:, 0::2] = np.sin(positions * div_term)
    pos_enc[:, 1::2] = np.cos(positions * div_term)

    return torch.tensor(pos_enc, dtype=torch.float32)

def learnable_positional_encoding(seq_len, d_model):
    return torch.nn.Parameter(torch.randn(seq_len, d_model))

# Example with sequence of 6 tokens and embedding size of 8
seq_len = 6
embedding_size = 8

sinusoidal_pe = sinusoidal_positional_encoding(seq_len, embedding_size)
learnable_pe = learnable_positional_encoding(seq_len, embedding_size)

print("Sinusoidal Positional Encoding:")
print(sinusoidal_pe)

print("\nLearnable Positional Encoding (Randomly Initialized):")
print(learnable_pe)
```

Sinusoidal Positional Encoding:

```
tensor([[ 0.0000e+00,  1.0000e+00,  0.0000e+00,  1.0000e+00,  0.0000e+00,
          1.0000e+00,  0.0000e+00,  1.0000e+00],
        [ 8.4147e-01,  5.4030e-01,  9.9833e-02,  9.9500e-01,  9.9998e-03,
          9.9995e-01,  1.0000e-03,  1.0000e+00],
        [ 9.0930e-01, -4.1615e-01,  1.9867e-01,  9.8007e-01,  1.9999e-02,
          9.9980e-01,  2.0000e-03,  1.0000e+00],
        [ 1.4112e-01, -9.8999e-01,  2.9552e-01,  9.5534e-01,  2.9996e-02,
          9.9955e-01,  3.0000e-03,  1.0000e+00],
        [-7.5680e-01, -6.5364e-01,  3.8942e-01,  9.2106e-01,  3.9989e-02,
          9.9920e-01,  4.0000e-03,  9.9999e-01],
        [-9.5892e-01,  2.8366e-01,  4.7943e-01,  8.7758e-01,  4.9979e-02,
          9.9875e-01,  5.0000e-03,  9.9999e-01]])
```

Learnable Positional Encoding (Randomly Initialized):

Parameter containing:

```
tensor([[-0.6954, -0.2330, -0.3798, -0.5834, -0.4723, -1.2842, -0.5812,  0.7591],
        [-1.3863, -0.4606, -1.5583, -0.6430,  1.6454, -1.6722, -0.8262, -0.0435],
        [ 0.4604,  1.4819, -0.6303,  1.0177, -0.8603,  1.4300,  1.4896, -0.3780],
        [-0.2474,  0.2699, -0.2627, -0.5891,  0.9769,  0.8027,  0.1052,  0.4878],
        [-0.9591,  1.5292, -0.2796,  2.0687,  0.5212, -2.1484,  0.9244,  0.1710],
        [-1.5114, -1.3014, -1.8583, -0.1906,  1.1627,  2.7021,  1.0032, -0.1064]],
        requires_grad=True)
```

INFERENCE

The sinusoidal positional encoding provides a fixed, interpretable representation of positions in a sequence, ensuring meaningful relative positioning. The learnable positional encoding, being trainable, allows the model to learn optimal representations for position information, potentially improving performance in specific tasks.